

# CyDT: The Cyprus Digital Twin for Resilient Cities and Robust Critical Infrastructure Systems

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**Abstract**—Modern smart cities now place top priority on keeping Critical Infrastructure Systems (CIS) like power grids, water networks, transportation systems, and telecommunications both resilient and highly efficient. Recent advances in Digital Twin (DT) technology have made it an attractive tool for managing these systems by generating dynamic, data-driven virtual models of cyber-physical assets and their interconnections. Despite widespread adoption of DTs across many industries, there is no risk-based, scenario-driven DT framework readily available to researchers, smart city operators, and CIS managers for bolstering urban resilience. To fill this gap, we present the Cyprus Digital Twin (CyDT), a one-of-a-kind digital platform designed to manage and optimize interdependent CIS. We detail CyDT's modular architecture outlining its principal components and spotlighting its novel features and capabilities and demonstrate its adaptability and effectiveness in real-life use cases that conduct risk-based resilience analyses of power and water systems.

**Index Terms**—Digital Twin; Critical Infrastructures; Resilience

## I. INTRODUCTION

As urban populations grow and cities become increasingly complex, ensuring the resilience and efficient operation of Critical Infrastructure Systems (CIS), such as power grids, water networks, transportation systems, and telecommunications, has become a top priority for modern smart cities. Recent advancements in Digital Twin (DT) technologies make them increasingly popular for managing such systems by creating dynamic, data-driven virtual replicas of physical assets and their interactions.

Previous works on smart city DTs have demonstrated how semantic models combined with Machine Learning (ML) and Artificial Intelligence (AI) enhance situational awareness, event detection, and decision-making [1]. Beyond operational optimization, DTs are increasingly being leveraged for cybersecurity applications within CIS, enabling threat simulations and secure-by-design architectures [2]. Authors in [3] provide a broad overview of global DT initiatives, emphasizing both the opportunities and integration challenges that still exist. A recent work explores the benefits of AI-driven DTs in

enhancing the resilience of CIS, highlighting their potential to mitigate natural disasters and cybersecurity threats [4].

Authors in [5] demonstrate the applicability of DTs in urban planning, where the DT provides data on congestion, pollution, flooding, and building simulations, making urban planning decisions much more transparent and informed. Complementary to this, authors in [6] introduce the concept of the Social Urban Digital Twin (SUDT), which integrates both the physical and social environments of a city.

Authors in [7] provide a comprehensive overview of the applications of DTs in transportation and power systems. In transportation, DTs enable real-time analysis of traffic and pedestrian behavior, leading to better decision-making, enhanced efficiency, mental health and well-being. In the energy sector, DTs for microgrids and power grids enhance stability, resilience, and fault detection, while supporting accurate forecasting and system optimization. The work in [8] focuses on False Data Injection Attacks (FDIA) in smart grids, particularly in power systems state estimation. They demonstrate the potential of DTs integrated with Recurrent Neural Networks (RNN) to detect FDIA by leveraging temporal and spatial correlations in data. Considering interconnected transportation-power CIS, a DT is proposed in [9] that focuses on transportation disruptions due to power outages caused by extreme weather events, e.g., the 2021 Texas power crisis. Recent works focus on disaster risk management [10], [11].

These DT platforms do not address joint risk and impact assessment of incidents and threats together with possible recovery actions. To this end, the main questions that we aim to answer are: i) what is the *risk* and *impact* of specific incidents and service disruptions to the city and associated CIS and ii) which *interventions* and *corrective actions* can be taken to recover? A risk-based scenario-driven DT would help answer both questions improving the *resilience* of smart cities and CIS; yet, such a DT is not yet available. In this work, we introduce the Cyprus Digital Twin (CyDT)<sup>1</sup>, which is a unique digital platform to assist with the management and operation of interdependent CIS. The contributions of this work are twofold:

- We present in detail the modular architecture of the CyDT from a software engineering viewpoint describing the

<sup>1</sup>Cyprus Digital Twin (CyDT) overview: <https://youtu.be/86WFkxe5fg4>

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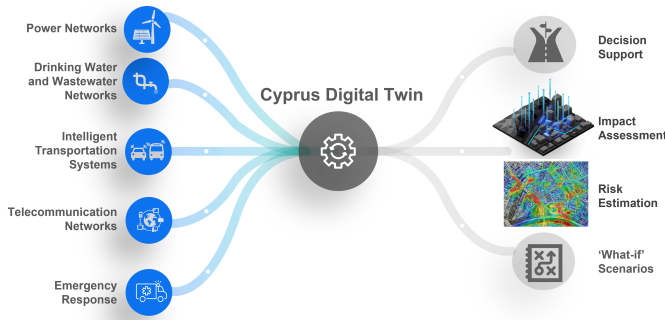


Fig. 1. Cyprus Digital Twin concept.

main building blocks and focusing on the platform's novel and distinctive features and functionalities;

- We showcase the flexibility and effectiveness of CyDT through representative use case scenarios that offer an informative perspective for risk-based analysis of the resilience of power and water CIS.

The rest of the paper is structured as follows. Section II outlines the CyDT system architecture and building blocks. Section III presents how these elements work in tandem to create the envisioned digital replica of Cyprus offering powerful features and functionalities. Section IV demonstrates how the integrated tools and data available in the CyDT are instantiated to implement real-life scenarios addressing different aspects of resilience. Finally, Section V provides concluding remarks and directions for future work.

## II. SYSTEM ARCHITECTURE

### A. Overview of the CyDT

The CyDT considers various CIS (e.g., power networks, drinking water and wastewater networks, intelligent transportation systems, and telecommunication networks), and the operations of first responders during emergency situations, as shown in Fig. 1. This DT platform offers high-resolution emulation of the actual operation of CIS within the urban and sub-urban environment by incorporating well-established libraries and software for simulation (e.g., EPANET for water networks, SUMO for transportation and mobility, Matlab, Python, etc.) and by modeling the interdependencies and cascading effects among CIS; all seamlessly integrated in the CyDT backend.

An intuitive and user-friendly interface enables exploration of complex use cases to investigate and answer questions related to monitoring, management, control, security, and resilience of CIS under varying conditions. Users are able to create complex 'what-if' scenarios through workflows (e.g., based on Node-RED) and execute them, to perform risk estimation and impact assessment in a fully parameterized and easily configurable virtual environment. These features and the overall CyDT architecture meet the requirements collected by various stakeholders (i.e., end-users of the platform) including policy makers, CIS operators, Smart City managers, and emergency response organizations to assist them in their daily operations and decision-making process.

### B. Main Building Blocks

The high-level architecture of the CyDT is illustrated in Fig. 2 and comprises the following building blocks.

1) *Data Sources*: A wide variety of input data sources is considered to ensure comprehensive coverage of diverse CIS domains. These include IoT devices deployed in the field that provide real-time sensor data, and testbeds developed at the KIOS Center of Excellence, University of Cyprus (KIOS CoE). Additional sources include national and EU open data repositories publishing structured datasets across sectors like mobility, energy, and environment, as well as Geographic Information System (GIS) data including OpenStreetMap for spatial and infrastructural analysis. The Cyprus National Access Point (CyNAP)<sup>2</sup> serves as a centralized platform for accessing transportation-related data, while private datasets from collaborating organizations contain specialized or proprietary information not publicly available.

2) *Data Collection*: The CyDT harvests different kinds of CIS-related data including: i) historical, statistical, and raw measurement data, e.g., data from distributed IoT devices collected through the KIOS IoT Platform, ii) data of various types, e.g., static, semi-static, and dynamic, and iii) data of different formats, e.g., structured or unstructured, streaming or in batches. In addition, flexible data collection interfaces are available including data connectors and endpoints (e.g., for real-time and lightweight data streaming through the MQTT protocol), message brokers, Apache Kafka for high-frequency streaming data, and HTTP collectors that retrieve data from web-based APIs and open data portals.

3) *Data Warehouse*: Collected data is managed within an integrated Data Warehouse. This process is orchestrated by the Data Management System (DMS) that organizes and structures the data for efficient access and use, while the Data Lake stores the multi-source and heterogeneous datasets described previously, as well as models (e.g., graph models for infrastructure networks) and scenarios (e.g., descriptions, workflows, and parameter configurations). State-of-practice technologies are used for data storage (e.g., PostgreSQL for relational data, PostGIS for geospatial data, InfluxDB for time series data) and archiving (e.g., OpenHistorian) in the CyDT Data Lake. Additionally, a GeoServer is linked to support the visualization, distribution, and processing of spatial data.

4) *Digital Twin Platform*: At the core of the system is a set of integrated services that enable seamless data interaction, processing, and decision-making. RESTful APIs facilitate smooth communication between components and external applications, while a backend system manages essential functionalities. A dedicated Services Management System, incorporating a Celery Orchestrator, RabbitMQ for task queuing, and Celery Workers for distributed task execution, ensures reliable and scalable service coordination. In addition, a Services Repository hosts a variety of components including simulators, predictive analytics algorithms, optimization solvers, and analytics tools that generate actionable insights.

<sup>2</sup>Cyprus National Access Point, <https://www.traffic4cyprus.org.cy/>

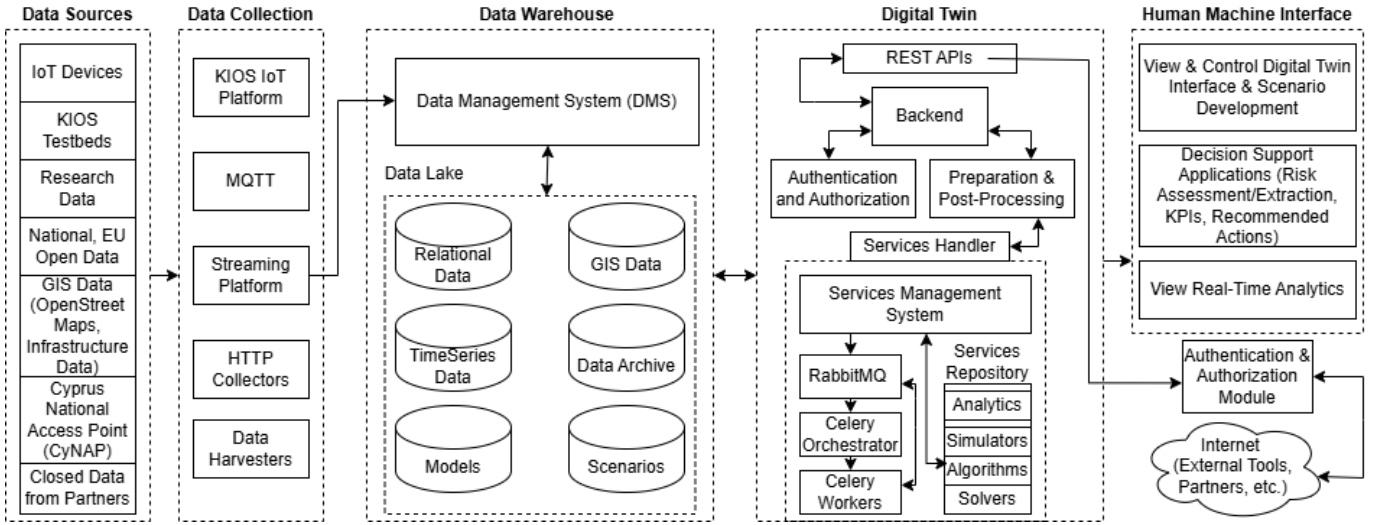


Fig. 2. Cyprus Digital Twin high-level architecture with the main building blocks.

System security is maintained through an Authentication and Authorization Module, which ensures that only verified users and systems can interact with restricted-access resources.

5) *Human-Machine Interface*: Users interact with the system through a Human-Machine Interface (HMI) that facilitates operational oversight and control of the CyDT. This interface enables scenario development and simulation, allowing users to model and assess the behavior of interconnected CIS under varying conditions. Decision Support Applications provide tools for risk assessment and the evaluation of Key Performance Indicators (KPI), supporting informed decision-making. Additionally, Real-Time Analytics components offer continuous monitoring and visualization of system performance, enhancing situational awareness and responsiveness.

### III. MAIN FEATURES AND FUNCTIONALITIES

The CyDT offers several features such as dataset visualization, ‘what-if’ scenario creation, interactive tools for analysis and decision support and APIs for exchanging data and interacting with other Digital Twin applications and services.

#### A. Data Visualization

Standalone data often lack context on its own and visualization plays a crucial role in revealing trends or even patterns, which may otherwise go unnoticed. To this end, the CyDT offers several views for enabling users to efficiently interact with and better understand the data.

The *Interactive map of Cyprus* interface enables users to view spatial data categorized by domain. These data go through a preprocessing stage that converts them into GeoJSON format and subsequently they are cached with *Redis*. This approach guarantees that users can select the desired dataset with minimal latency. The interactive map of Cyprus displaying the national forests dataset is illustrated in Fig. 3a. On the top right corner, there is a multi-layered control which organizes the datasets into different domains such as

Transportation, Water and Power while each dataset contains either points or shapes. When a dataset is selected, it is rendered on the map. These datasets are static or semi-static, i.e., they are usually refreshed on a monthly or weekly basis.

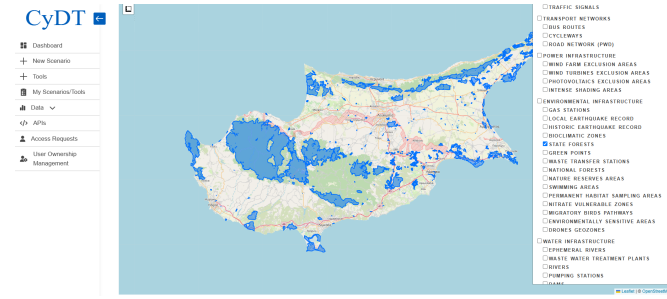
Besides (semi-)static datasets, the CyDT facilitates the visualization of live sensory data, e.g., information related to public transportation, air quality, and weather. For instance, the route of a selected bus (blue line), bus stops along the route (red dots), and telemetry data (bearing and speed) is illustrated in Fig. 3b. Data is refreshed every 10 to 15 seconds allowing users to track bus movements in real time. The CyDT stores up to one year of historical data for offline analysis and replaying data for replicating previous experiments.

On the other hand, air quality and weather data are visualized through a dedicated *Grafana* interface integrated into the CyDT. This interface presents multi-dimensional air and environmental quality data and weather information in an accessible and interactive way through intuitive visualizations like graphs and time-series charts. This helps users easily identify trends, track changes over time, and quickly access data pertaining to different locations. Additionally, the dashboard provides real-time alerts, ensuring users stay informed and can respond to environmental changes as they happen.

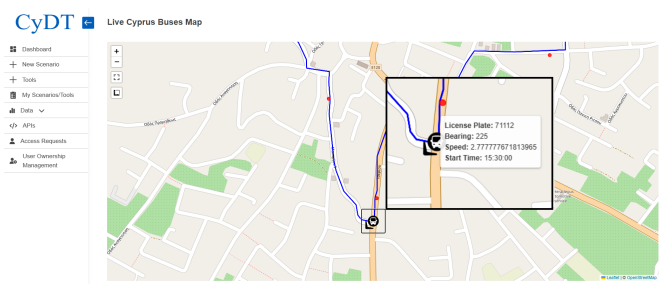
While Grafana excels at displaying real-time sensor data, the platform also harvests statistical data from the Statistical Service of Cyprus [12], offering additional insights. These datasets are accessed through *GraphQL*, a flexible and efficient data query language, which is fully integrated in the CyDT, enabling users to tailor their data requests directly through the system. For example, users can extract specific data such as the population of a city, including any age group or specific age range, and feed it into a use case scenario.

#### B. User Dashboard

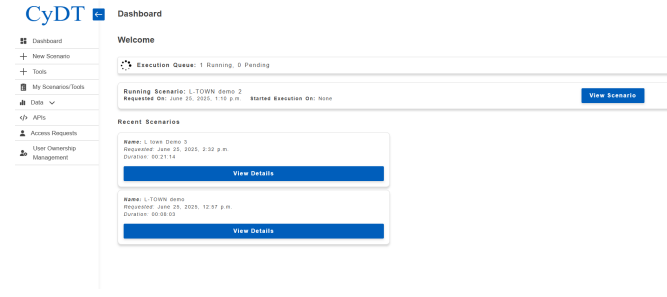
The CyDT offers a user-friendly interface designed to simplify interactions and enhance user experience. Users can



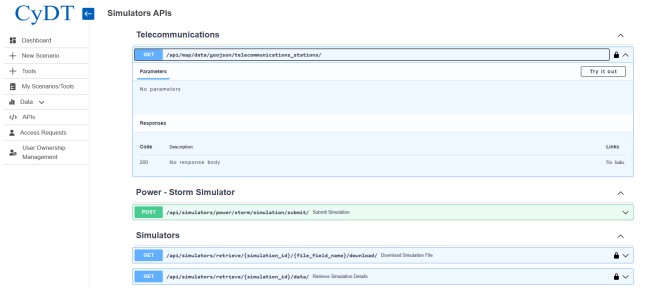
(a) Interactive map of Cyprus displaying the selected national forests dataset



(b) Live map showing the route and telemetry data of a bus



(c) User dashboard for tools instantiation and scenario creation



(d) Subset of APIs for cross-platform interconnection

Fig. 3. Key features of the Cyprus Digital Twin (CyDT) platform, including data visualization (a–b), user dashboard (c), and system API endpoints (d).

see the progress of their work, e.g., which scenarios are still pending for execution, running, and which have finished, as shown in Fig. 3c. The navigation bar on the left side provides access to the various features of the CyDT.

For instance, users can develop and execute scenarios for the first time by selecting the *New Scenario* tab, choosing the desired scenario from a list of CIS categories (i.e., power, water, etc.), and modifying the scenario parameters to meet their needs. For each scenario there is a predefined visualization view that displays the results, so that the users can revisit their strategy or decision and adjust the parameters as necessary to re-run the scenario. Similarly, the *Tools* tab, provides access to various tools and simulators (e.g., wind simulator, pipe length calculator, water leakage detection algorithm, etc.) that can be instantiated as part of a larger scenario. The *My Scenarios/Tools* tab provides a convenient list-view of previously created scenarios and configured tools to facilitate multiple consecutive runs.

The *Data* tab provides access to the CyDT data inventory, including easy search with multi-category criteria, and visualizes datasets as described in Section III-A. Lastly, the *APIs* tab, which is discussed in the next section, lists all endpoints available to the user for exchanging information with external services/applications or other DTs.

### C. API-based Interaction with External Entities through

All datasets within the platform can be conveniently accessed and extracted via *API calls*, providing users with the flexibility to use the data for their own applications or analysis. These APIs allow users to retrieve specific datasets, whether for real-time monitoring or historical analysis. For instance,

users can easily integrate data into their own systems or workflows, enabling further customization and personalized use based on their specific needs. Beyond data provision, the CyDT supports API access to other features including tools and scenarios. For instance, users can remotely instantiate the required datasets (even subsets of data using GraphQL), configure the target simulator by passing through the parameters with the proper API call, trigger the envisioned scenario, and finally collect the results for visualizing and/or further processing with proprietary tools. A subset of the available API endpoints is illustrated in Fig. 3d.

### D. Data Access and User Ownership Management

Since the CyDT hosts both *public* and *confidential* datasets, an access-control mechanism is foreseen to manage user access to restricted data. Essentially, users can submit *Access Requests* for confidential datasets by providing a justification. These requests are reviewed by the CyDT admin team or the dataset owner, who can either approve or reject them. This ensures that access to sensitive data is controlled and aligned with data governance policies in place.

The *User Ownership Management* functionality is designed to assign users as owners of specific datasets, scenarios, or tools within the CyDT. This feature is accessible only to admin users to ensure controlled and consistent assignment of ownership roles. By clearly defining ownership, the system streamlines the access request process, allowing the appropriate individual to review and respond to requests efficiently. This mechanism enhances accountability and ensures that permissions are managed by the most relevant users, reducing administrative overhead for system-wide access control.

#### IV. USE CASE SCENARIOS

In this section, we demonstrate the CyDT in representative scenarios related to the resilience of CIS under various conditions. For each scenario we provide the parameters used along with the visual output produced after execution.

##### A. Use Case 1: Resilience of Power Lines to Windstorms

As the frequency and intensity of extreme weather events escalate due to climate change, power systems are becoming increasingly susceptible to disruptions. Windstorms, in particular, pose a significant threat by damaging transmission and distribution infrastructure, leading to prolonged outages and substantial recovery costs.

To address this challenge, the CyDT platform incorporates a specialized scenario for assessing the resilience of power lines to windstorms. This use case enables the modeling and simulation of windstorms with dynamic spatial and temporal characteristics and their interaction with the power grid. It integrates historical meteorological data with synthetic windstorm patterns to simulate storm progression and assess the potential impact on power network assets. The primary objective is to quantify the effects of storms on the grid and provide a visual tool for identifying vulnerable components, thereby enhancing preparedness and mitigation strategies.

Figure 4a illustrates the simulation output of an illustrative scenario. It presents a synthetic power grid topology in the east coast of Cyprus close to Larnaca International Airport over which a simulated windstorm—depicted as a moving circle—propagates through the network. As the storm intersects various parts of the grid, lines at risk of failure are highlighted in red, offering an intuitive view of potential vulnerabilities and the progression of the event. While not based on real infrastructure data, the example demonstrates CyDT's analytical capabilities. The underlying framework is fully customizable: users can upload actual network topologies and define storm parameters—such as gust wind speed, storm intensity, duration, and trajectory—to simulate realistic, location-specific scenarios. The fragility curve used is based on [13], and the vulnerability assessment approach aligns with methodologies such as those proposed in [14]. Input parameters for this scenario are summarized in Table I.

We present a complex multi-sector scenario where a storm affects the power grid creating cascading effects to the interconnected cellular communication network in [15].

##### B. Use Case 2: Robustness of Water Distribution Systems to New Developments

This scenario enables users to evaluate the impact of new infrastructure or demand changes within an urban District Metered Area (DMA). Users can select a DMA, specify a node of interest, and define minimum and maximum thresholds for both demand and pressure, as well as the number of simulations to run. The scenario takes as input a water network model, a node ID, demand and pressure thresholds, the number of simulations, and the uncertainty of the base demand of the nodes. It performs Monte Carlo simulations by varying the

demand of the specified node and outputs the minimum and maximum pressures across all nodes for all simulations. It also generates a report identifying any nodes and times when the pressure falls below or exceeds the specified constraints. The report includes the current and future values of daily demand for the specified DMA, as well as the demand and pressure at the times of minimum and maximum pressure. The scenario utilizes the EPANET-Python Toolkit (EPyT), an open-source software developed by the KIOS CoE [16]. EPyT provides a programming interface in Python for EPANET, which is a hydraulic and water quality modeling software created by the US Environmental Protection Agency (EPA).

We analyze the scenario output in Fig. 4b following the introduction of a new development at node  $n_{25}$  with multiple simulations (scenario parameters are listed in Table II). This development represents a localized increase in demand and is used to evaluate the network's response under modified conditions. The output is a map of the water distribution network; dark blue nodes represent junctions where pressure levels fall below the minimum acceptable threshold, whereas light blue nodes denote junctions receiving pressure within the acceptable range. The low-pressure nodes indicate areas where service continuity is at risk, providing valuable insights into potential issues introduced by the new development.

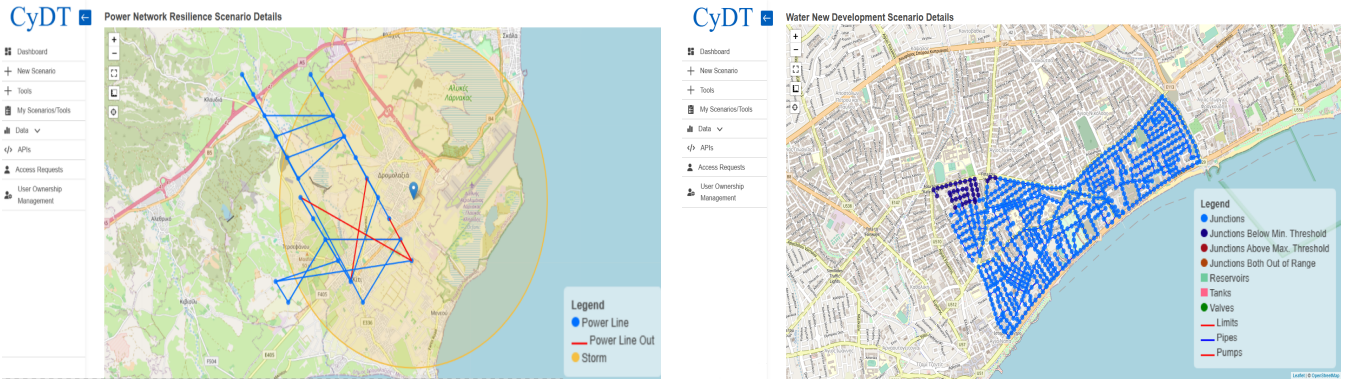
#### V. CONCLUSIONS

This work presented the design and implementation of CyDT, a comprehensive digital platform tailored to enhance the resilience and operational understanding of interdependent CIS within urban environments. The CyDT facilitates the exploration of complex what-if scenarios that are essential for risk assessment and informed decision-making. The architecture of CyDT brings together a modular backend with flexible data pipelines, scalable simulation frameworks, and interactive dashboards, allowing stakeholders to model, monitor, and evaluate the behavior of key CIS under both routine and abnormal conditions. The platform's ability to visualize spatial and temporal data, coupled with its API-driven interoperability, ensures adaptability across domains and use cases. Through two representative scenarios, namely resilience of power networks to windstorms and robustness of water systems to urban developments, we demonstrated how CyDT can support risk-based proactive planning. These examples emphasize the platform's potential to assist policymakers, city operators, and CIS managers in anticipating hazard events and subsequent cascading effects; thus, avoiding service disruption and ensuring business continuity.

Future work will focus on extending the CyDT with real-life scenarios from additional CIS sectors, e.g., transportation, enhancing risk analysis and impact assessment through new tools and interactive user interfaces, and analysis of rapid response strategies.

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(a) Windstorm (circle) passing over power grid (blue) damaging lines (red) (b) Low (dark blue) or nominal (light blue) pressure due to new development

Fig. 4. Representative scenarios in the CyDT for analysing the resilience of CIS, including the power grid (a) and water distribution networks (b).

TABLE I  
INPUT PARAMETERS FOR THE SCENARIO ON THE RESILIENCE OF POWER LINES TO WINDSTORMS

| Parameter                          | Value(s)     | Description                                                                               |
|------------------------------------|--------------|-------------------------------------------------------------------------------------------|
| Mean (Fragility Curve)             | 3.8          | Mean of the lognormal fragility function used to model wind-induced line failures.        |
| Gust Wind Speed Range (km/h)       | 40–220       | Range of gust wind speeds used in the simulation.                                         |
| Storm Heading Range (° from North) | 20–160       | Clockwise heading range defining storm trajectory; recommended to exceed 30° for realism. |
| Initial Storm Radius Range (km)    | 1–5          | Minimum and maximum radius of the windstorm at its western boundary.                      |
| Line Restoration Time Range (h)    | 3–11         | Time range assumed for restoring damaged power lines.                                     |
| Wind Speed Region Range (km/h)     | 11.34–16.776 | Min and max wind speeds across the simulation region.                                     |
| Number of Simulations              | 1            | Number of storm events simulated.                                                         |

TABLE II  
INPUT PARAMETERS FOR THE SCENARIO ON THE WATER NETWORK RESILIENCE UPON NEW DEVELOPMENT

| Parameter                      | Value | Description                                                   |
|--------------------------------|-------|---------------------------------------------------------------|
| Node ID                        | n25   | Identifier for the specific node in the network.              |
| Minimum Demand (CMH)           | 0     | Minimum demand assigned to the node.                          |
| Maximum Demand (CMH)           | 5     | Maximum demand assigned to the node.                          |
| Minimum Pressure Threshold (m) | 30    | Minimum pressure threshold below which service is inadequate. |
| Maximum Pressure Threshold (m) | 90    | Maximum pressure limit to ensure system safety.               |
| Number of Simulations          | 5     | Number of simulations run using these parameters.             |

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